

MATHS

SECTION-A

(3) 3

(3) $\sqrt{3} \tan 60$

(4) 100°

(4) ~~area~~ (8 cm)

~~(4) ~~area~~ (a=0)~~

(4) (Four de.)

(3) 341

(2) (0, -1)

(4) No solⁿ

(1) $-\frac{9}{2}$

Q11 (1) $k \neq -3$

Q12 (1) 8

Q13 (1) Similar

Q14 (1) $x^3y^4z^4$

Q15 (3)

Q16 (3)

Q17 (1)

Q18 (2)

Q19 (3)

Q20 (1)

SECTION-B

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$$

$$30 \times 12 \times 2 + 5$$

$$= 60 \times 12 + 5$$

$$= 725$$

$$= \text{which is divisible by 5}$$

4 for infinite = $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$$\frac{4}{k} = \frac{6}{18} = \frac{2}{6} = \frac{1}{3}$$

$$\boxed{k=12}$$

(0,0)

A

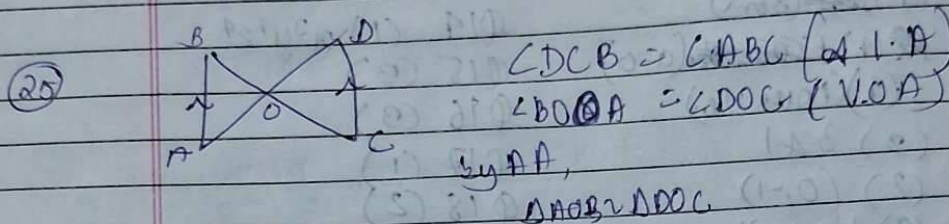
B

(6,8)

$$\sqrt{(-6-0)^2 + (8-0)^2}$$

$$= \sqrt{36 + 64} = \sqrt{100} = 10 \text{ units}$$

(24) $3A = \cos(A - 10^\circ)$



SECTION-6

(26) $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}$

$$\frac{\sin^2 \theta + (1 + \cos \theta)^2}{(1 + \cos \theta)(\sin \theta)}$$

$$= \frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)} = \frac{\sin^2 \theta + \sin^2 \theta + \cos^2 \theta + \cos^2 \theta + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)}$$

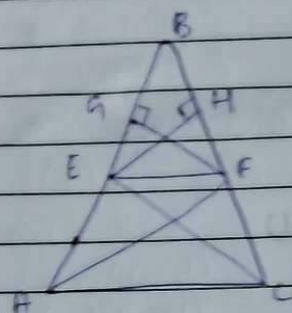
$$= \frac{\sin^2 \theta + \cos^2 \theta + 1 + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)} = \frac{2 \sin^2 \theta + 2 \cos^2 \theta + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)}$$

$$= \frac{1 + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)} = \frac{2 \sin^2 \theta + 2 \cos^2 \theta + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)} = \frac{2(1) + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)}$$

$$= \frac{2(1+\sqrt{3})}{(1+\sqrt{3})(2\sqrt{3})} = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{1 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{\sqrt{3}}{3}$$

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$$\text{area of } \Delta BEF = \frac{1}{2} \times BE \times BF$$

$$\text{area of } \Delta AEF = \frac{1}{2} \times EF \times AF$$

$$\frac{\text{area of } \Delta BEF}{\text{area of } \Delta AEF} = \frac{BE}{EF}$$

$$\text{and } \frac{\text{area of } \Delta BEF}{\text{area of } \Delta CEF} = \frac{BF}{FC}$$

by property of 11th theorem the same
both are equal area.

$$\text{area of } \Delta CEF = \text{area of } \Delta AEF$$

$$\therefore \frac{BE}{EF} = \frac{BF}{FC}$$

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present age of A = x
present age of B = y

$$(x-5) = 2(y-5)$$

$$x-5 = 2y-10$$

$$x-2y = -10+5$$

$$x-2y = -5 \quad \text{--- (i)}$$

$$(x+10) = 2(y+10)$$

$$x+10 = 2y+20$$

$$x-y = 10 \quad \text{--- (ii)}$$

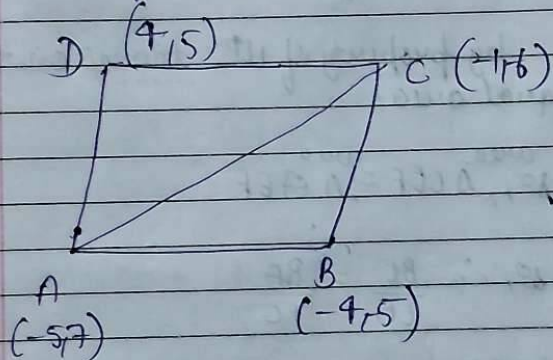
$$\begin{aligned} x - 2y &= -5 \\ -x + y &= -10 \\ \hline y &= -20 \\ y &= 20 \end{aligned}$$

means. Puri age is 20

$$\begin{aligned} \text{Sagar age} &= x - 20 = 10 \\ &= \boxed{x = 30} \end{aligned}$$

and when Puri was born, Sagar was of 10 years.

29.4



$$\text{area of } \frac{1}{2} [x_3(y_2 - y_1) + x_2(y_3 - y_1) + x_1(y_3 - y_2)]$$

~~Not in syllabus~~
Not in school syllabus.

$$\begin{array}{r} 304 \quad 3 \overline{) 861} \\ \underline{72} \\ 141 \\ \underline{120} \\ 21 \\ \underline{21} \\ 0 \end{array} \quad \begin{array}{r} 3 \overline{) 353} \\ \underline{30} \\ 53 \\ \underline{45} \\ 81 \\ \underline{78} \\ 31 \\ \underline{30} \\ 1 \end{array} \quad \begin{array}{r} 3 \overline{) 91} \\ \underline{6} \\ 31 \\ \underline{30} \\ 1 \end{array}$$

= 123 ncr.

31.)

$$(5, -2) \quad (n, 0) \quad (-3, 2)$$

$$n = \frac{-3+5}{2}$$

$$= \boxed{n=1} = (1, 0)$$

32.)

$$\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \csc \theta$$

$$= \frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}}$$

$$= \frac{\frac{\sin \theta}{\cos \theta}}{\frac{\sin \theta - \cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{\frac{\cos \theta - \sin \theta}{\cos \theta}}$$

$$= \frac{\sin \theta}{\cos \theta} \times \frac{\sin \theta}{\sin \theta - \cos \theta} + \frac{\cos \theta}{\sin \theta} \times \frac{\cos \theta}{\cos \theta - \sin \theta}$$

$$= \frac{\sin^2 \theta}{(\cos \theta)(\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)}$$

$$= \frac{\cancel{\sin^2 \theta}}{\cancel{\sin \theta} \cdot \cos \theta - \cos^2 \theta} + \frac{\cancel{\cos^2 \theta}}{\sin \theta \cdot \cancel{\cos \theta} - \sin^2 \theta}$$

$$\frac{\sin^2}{\cos(\sin-\cos)} + \frac{\cos^2}{\sin(\cos-\sin)}$$

$$= \frac{\sin^2(\sin)(\cancel{\cos-\sin}) + \cos^2(\cos)(\cancel{\sin-\cos})}{(\cos)(\sin-\cos)(\cancel{\sin}) (\cancel{\cos-\sin})}$$

$$= \frac{\sin^2 + \cos^2}{\cos + \sin}$$

$$= \frac{1}{\sin + \cos}$$

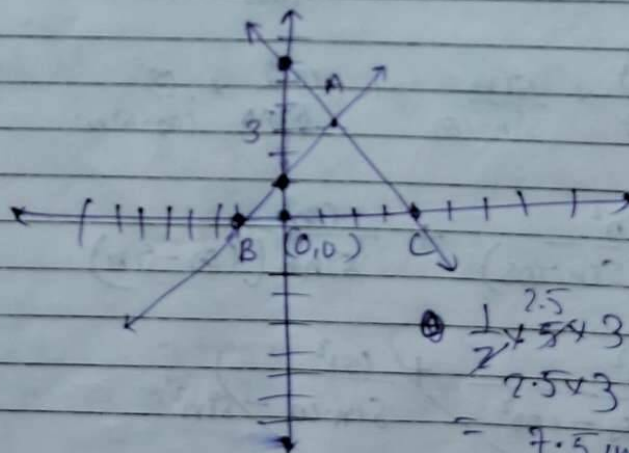
(33)

$$x - y = -1$$

$$\begin{array}{c|c|c} x & 0 & -1 \\ \hline y & +1 & 0 \end{array}$$

$$3x + 2y = 12$$

$$\begin{array}{c|c|c} x & 0 & 4 \\ \hline y & 6 & 0 \end{array}$$



$$\begin{aligned} & \frac{1}{2} \times 2.5 \times 3 \\ &= 7.5 \text{ unit}^2 \end{aligned}$$

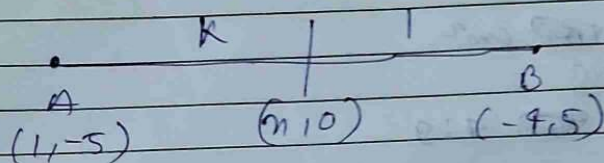
344

$$2x^3 - 4x - x^2 + 2$$

$$2x^2 - x^2 - 4x + 2$$

$$x = \sqrt{2}, x = -\sqrt{2} \text{ and } x = ?$$

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$$x = \frac{k(-4) + 1}{k+1}, \quad 0 = \frac{5k + (-5)}{k+1}$$

$$x = \frac{-4k + 1}{k+1} = 0 = \frac{5k - 5}{k+1}$$

$$0 = (k-1)$$

$$= \boxed{k=1}$$

$$x = \frac{-4(1) + 1}{1+1}$$

$$x = \frac{-4+1}{2}$$

$$\boxed{x = -\frac{3}{2}}$$

point $\left(-\frac{3}{2}, 0\right)$

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$$(1) (2) 0, 12$$

$$(11) (9) 3, 5$$

$$(111)$$



$$\sqrt{(6-3)^2 + (9-5)^2} = \sqrt{3^2 + 4^2} = 5 \text{ units}$$

~~(12)~~

37.7 (f) (3) Parabola

(i) (3) 0

(ii) (3) -3/2

38.4 (f) (1) 48 cm²

(i) (1) RHS

(ii) (2) 4:9

$$(2) + (2) = 0$$

$$\frac{1 + (1) + 1}{1 + 1} = 0$$

$$2 - 1 = 0$$

$$(1 - 1) = 0$$

$$[1 - 1] =$$

$$1 + 1 + 1 = 0$$

$$(1 + 1)$$

$$1 + 1 + 1 = 0$$

$$(1 + 1)$$

$$1 + 1 + 1 = 0$$

$$1$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$1 + 1 + 1 = 0$$

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